Nicholas Eisenberg, Ph.D.

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Experience

Senior Scientist - Machine Learning *June 2022 - Present*
Nevada National Security Site

- **Retrieval-Augmented Generation (RAG) System for Internal Knowledge Base**: Designed and deployed a RAG system over internal Python/C++ codebases and company documents. Parsed source code using `AST`, `jedi`, and `clang.cindex` to create structured chunks, encoded them with `all-MiniLM-L6-v2`, and indexed them using `hnswlib` for fast vector search. Served the model with `vLLM` and integrated the system into Open WebUI for accessible querying.
- **Image Denoising with Diffusion**: Trained a denoising diffusion probabilistic model (DDPM) to clean up scientific images with noise artifacts. Tuned model architecture and noise schedules to achieve effective overfitting and generalization.
- **Vehicle Detection Using YOLOv5**: Developed a YOLOv5-based object detection pipeline for a pattern-of-life project targeting vehicle detection. Fine-tuned model and deployed it on a Jetson Nano for real-time inference.
- **Neo4j Interface with Django**: Built and deployed a Django application enabling non-technical users to manage the internal Neo4j knowledge graph. Containerized the system using Docker and Docker Compose for reproducible deployment across environments.
- **Maintainer of Internal Monolithic Repo**: Maintain and evolve a large-scale internal monorepo, enforcing development workflows, resolving dependency issues, and improving CI/CD practices.

Technical Skills

Programming Languages: Python, Bash, Lua, SQL, Cypher, LaTeX
Libraries: PyTorch, Pandas, Scikit-Learn, Matplotlib, Plotly, Django, Streamlit, Sqlalchemy
Developer Tools: Git, Linux, Vim, Tmux, AWS, SLURM, Docker, Neo4j

Personal Projects

iron.nvim Maintainer (GitHub link)
iron.nvim is an interactive and cross operating system REPL plugin for neovim, written in Lua, that supports many interpretted languages such as Python, R, Lua and more. My roles as maintainer inlude ensuring the health of the code base, adding new features and responding to ongoing issues.

Education

Auburn University - Ph.D. *June 2022*
Research area: Stochastic Partial Differential Equations
Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S. *December 2015 / January 2018*
Major: Applied Mathematics

Publications

1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). https://doi.org/10.1007/s40072-022-00258-6

2: Eisenberg, N., Chen, L. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). https://doi.org/10.1007/s10959-023-01302-4